A Comprehensive Literature Survey on a Deep Learning Based Method for Parkinson's Disease Detection Using Dynamic Features of Speech

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1 Introduction and Research Context

1.1 Background

Parkinson's Disease (PD) is currently the second most frequent neurodegenerative disease, after Alzheimer's disease. The disease manifests through two broad categories of symptoms: motor and non-motor symptoms.

Table 1: Major symptom categories associated with Parkinson's disease.

Symptom Category	Examples
Motor Symptoms	Tremor, bradykinesia (slowness of movement), rigidity (stiffness), and postural instability (poor balance)
Non-Motor Symptoms	Mood disorders, cognitive dysfunction, pain, sensory dysfunction, and dysautonomia

1.1.1 The Clinical Significance of Speech Analysis

Speech analysis has emerged as an important non-invasive approach for identifying Parkinson's disease. Motor speech disorders occur in more than half of PD patients and may manifest as very quiet speech, hurried speech, and reduced articulation precision.

Table 2: Clinical significance of speech analysis in Parkinson's disease.

Aspect	Detail
Prevalence	Motor speech disorders occur in more than half of PD patients.
Speech Disturbances	Very quiet speech, hurried speech, and reduced articulation precision.
Non-invasive	Speech analysis provides a non-invasive method for disease identification.
Early Detection	Voice changes can potentially be detected before disabling physical symptoms appear.
Clinical Impact	Earlier detection may enable timely intervention, potentially improving quality of life and reducing healthcare burden.

1.1.2 Established Acoustic Correlates of PD

Several acoustic measurements have been associated with Parkinson's disease.

Table 3: Established acoustic correlates of Parkinson's disease.

Measurement	PD Effect
Maximum Phonation Time	Shorter in PD patients
Jitter	Higher in PD patients
Shimmer	Higher in PD patients
Pitch Range	Decreased in PD patients
Phonation Threshold Pressure	Increased in PD patients

1.2 Limitations of Previous Approaches

1.2.1 Machine Learning Approaches

Traditional machine learning methods for Parkinson's disease detection generally rely on manually extracted acoustic features followed by statistical classification.

Table 4: Machine learning approaches and their limitations.

Category	Approach	Limitation
Traditional ML	Extract static features such as MFCC, pitch, jitter, and shimmer, followed by classification	Ignores temporal dynamics and relies heavily on feature engineering
Deep Learning (DL)	Hierarchical feature extraction	Many studies do not explicitly model interdependencies within sequential features

1.2.2 Dominant Traditional ML Methods

Table 5: Traditional machine learning methods used for PD detection.

Method	Studies
Support Vector Machine (SVM)	[12]–[14]
K-Nearest Neighbor (KNN)	[12]
Naive Bayes (NB)	[12], [15], [16]
Decision Tree	[12]
Genetic Algorithm	[14]
Ensemble Methods	[17], [18]

1.2.3 Traditional ML Pipeline

The conventional machine learning pipeline can be summarized as follows:

1. Extract global static features.
2. Perform feature selection using techniques such as LASSO, Relief, LLBFS, etc.
3. Perform dimensionality reduction, such as Principal Component Analysis (PCA).
4. Perform classification.

1.2.4 Deep Learning for PD Detection

Table 6: Deep learning approaches for Parkinson's disease detection.

DL Method	Studies	Focus
CNN	[19]–[22]	Feature-level and model-level combination; spectrogram analysis
End-to-End CNN	[20], [21]	Direct audio classification using time-frequency representations

1.3 The Gap: Dynamic Features

1.3.1 The Problem with Static Features

Static acoustic features summarize speech characteristics over an interval and therefore may lose important temporal information.

Table 7: Limitations of static speech features.

Issue	Explanation
Temporal Independence	Static features collapse measurements across time.
Loss of Sequential Information	Interdependencies in feature sequences are ignored.
Articulation Dynamics	PD affects speech articulation transitions, which are inherently temporal.
Clinical Relevance	Speech is a dynamic process, and static features may miss important temporal patterns.

1.3.2 Prior Evidence of Articulation Impairment in PD

Table 8: Previous evidence regarding articulation impairment in Parkinson's disease.

Study	**Finding**
Skodda & Visser (2011)	Vowel Articulation Index (VAI) is reduced in PD.
Rusz et al. (2013)	Monologue was found to be suitable for early PD detection, with up to 80% accuracy.
Arias-Vergara et al. (2017)	Continuous speech features involving prosody, intelligibility, and articulation improve discrimination.

1.3.3 Key Observation

When articulation transitions between healthy control (HC) speakers and PD patients are compared, differences can be observed in both the number of transitions and the fundamental frequency trajectory.

The number of articulation transitions is significantly different between HC and PD speakers. Furthermore, the trend of the fundamental frequency curve differs between the two groups.

A paired t-test produced:

$$p = 0.042 < 0.05$$

indicating that the observed difference was statistically significant under the reported analysis.

1.4 Research Objectives and Contributions

Table 9: Major contributions of the study.

Contribution	Significance
1. Articulation Transition Analysis	Demonstrated significant differences in transition count and F0 trend between HC and PD speakers.
2. Dynamic Feature Extraction	Measured energy content in unvoiced-to-voiced (onset) and voiced-to-unvoiced (offset) transitions.
3. Bidirectional LSTM Application	Reported as the first application of Bidirectional LSTM to dynamic articulation features for PD detection.
4. Comprehensive Comparison	Traditional ML and DL models were evaluated using two different evaluation methods.

1.4.1 Key Hypothesis

The central hypothesis is that PD patients produce abnormal unvoiced sounds and experience difficulty initiating and/or terminating vocal fold vibration. These abnormalities can potentially be observed by modeling the frequency content of unvoiced frames and transitions between voiced and unvoiced sounds.

2 Methodology

2.1 Dataset

The study uses the GYENNO dataset.

Table 10: GYENNO dataset characteristics.

Aspect	Detail
Total Subjects	45 (15 HC, 30 PD)
Gender	25 female, 20 male
Age Range	37–75 years
PD Stages	Hoehn & Yahr (HY) stages 1–5
Speech Tasks	Sustained monophonic /a/ ($\sim$5.5 s), short sentence ($\sim$5.5 s)
Total Samples	268
Recording Device	Smartphone
Sampling Rate	96 kHz
Feature Extraction	NeuroSpeech software
Language	Mandarin Chinese

2.2 Speech Features for PD Detection

2.2.1 Phonation Features

Table 11: Phonation features used for PD detection.

Feature	Description	PD Effect
Jitter	Temporal perturbation of fundamental frequency	Increased
Shimmer	Temporal perturbation of amplitude	Increased
APQ	Amplitude Perturbation Quotient	Increased
PPQ	Pitch Perturbation Quotient	Increased
F0 Contour	Fundamental frequency variation	Less stable in PD

Key Observation: The contour of the HC sample is more stable than the contour obtained from the PD patient for both monophonic /a/ and short sentence inputs.

2.2.2 Articulation Features

Table 12: Articulation features used for PD detection.

Feature	Description
VSA	Vowel Space Area
VPA	Vocal Pentagon Area
FCR	Formant Centralization Ratio
Onset	Energy in the unvoiced-to-voiced transition
Offset	Energy in the voiced-to-unvoiced transition

Key Observation: The sustained articulation analysis of monophonic /a/ shows that the contour of the HC speaker is more stable than that of the PD patient.

2.2.3 Dynamic Articulation Features

For continuous speech signals, articulation features are measured based on transitions between voiced and unvoiced segments.

Table 13: Dynamic articulation transition types.

Transition Type	Description
Onset	Energy in the transition from unvoiced to voiced segments
Offset	Energy in the transition from voiced to unvoiced segments

Each articulation transition is represented using 58 dynamic measurements.

Table 14: Composition of the 58 dynamic features.

Component	Count
BBEs (Bark Band Energies)	22
MFCCs (Mel-Frequency Cepstral Coefficients)	12
DMFCCs (First derivative of MFCCs)	12
DDMFCCs (Second derivative of MFCCs)	12
Total	**58**

2.3 Bidirectional LSTM Model Architecture

2.3.1 Why Bidirectional LSTM?

Different recurrent neural network architectures have different capabilities for modeling sequential speech information.

Table 15: Comparison of recurrent neural network architectures.

RNN Type	Advantage	Disadvantage
Simple RNN	Models sequential data	Vanishing gradient problem
LSTM	Overcomes vanishing-gradient problem and provides long-term memory	Only forward context
Bidirectional LSTM	Uses both forward and backward context	More complex and contains more parameters

2.3.2 Architecture Components

The overall architecture can be represented as:

```
Input: Dynamic Articulation Features (time series)
                |
                v
Forward LSTM Layer (from t=1 to N)
                |
                v
Backward LSTM Layer (from t=N to 1)
                |
                v
Fully Connected Layer
                |
                v
Output: PD or HC
```

2.3.3 LSTM Cell Equations

The LSTM cell uses forget, input, and output gates to control information flow.

Table 16: LSTM cell equations and their functions.

Gate	Equation	Function
Forget Gate	$f_t = \sigma(W_f x_t + W_{hf} h_{t-1} + W_{cf} c_{t-1} + b_f)$	Decides what information to forget
Input Gate	$i_t = \sigma(W_i x_t + W_{hi} h_{t-1} + W_{ci} c_{t-1} + b_i)$	Decides what information to store
Output Gate	$o_t = \sigma(W_o x_t + W_{ho} h_{t-1} + W_{co} c_t + b_o)$	Decides what information to output
Cell State	$c_t = f_t c_{t-1} + i_t \tanh(W_c x_t + W_{hc} h_{t-1} + b_c)$	Updates the memory state

2.3.4 Bidirectional LSTM Output

The Bidirectional LSTM computes a forward hidden sequence and a backward hidden sequence. These representations capture temporal information from both directions.

The forward hidden sequence can be represented as:

$$\overrightarrow{h_t}$$

and the backward hidden sequence as:

$$\overleftarrow{h_t}.$$

The two representations are combined to produce the output representation:

$$h_t = \left[\overrightarrow{h_t}; \overleftarrow{h_t}\right].$$

The final output can then be represented as:

$$y_t = W_y h_t + b_y.$$

2.3.5 Zero-Padding

All sequences of dynamic features are zero-padded to the same length before being fed into the network. This allows sequences with different numbers of articulation transitions to be processed using a common input representation.

2.4 Models Compared

2.4.1 Traditional ML Models (Static Features)

Table 17: Traditional machine learning models.

Model	Key Parameters
Decision Tree (DT)	Scikit-learn default settings
Multilayer Perceptron (MLP)	Hidden layers: 100, 200, 500 units
K-Nearest Neighbor (KNN)	$k = 3, 5, 10$
Gaussian Naïve Bayes (GNB)	Scikit-learn default settings
Support Vector Machine (SVM)	Linear, polynomial, RBF, and sigmoid kernels

2.4.2 DL Models with Dynamic Features

Table 18: Architecture of the end-to-end CNN.

Layer	Filters	Kernel Size	Pooling Size
Conv1	64	5	3
Conv2	128	3	3
Conv3	256	3	3

CNN (End-to-End) Additional CNN configuration:

- Dropout: 0.2 after each max-pooling layer.
- Activation functions: ReLU, Tanh, and Sigmoid tested separately.
- Input: Time-frequency representations including STFT, Mel spectrograms, and CQT spectrograms.

Table 19: RNN architectures evaluated in the study.

Architecture	Units	Dropout	Recurrent Dropout
LSTM	100	0.2	0.2
Bidirectional LSTM	100	0.2	0.2
Bidirectional GRU	100	0.2	0.2

RNN Models

2.5 Evaluation Methods

2.5.1 Method 1: 10-Fold Cross-Validation

Table 20: 10-fold cross-validation configuration.

Parameter	Value
Folds	10
Stratification	Preserve class distribution
Metrics	Accuracy, F-score, Specificity, Sensitivity, MCC

2.5.2 Method 2: Subject-Independent Split (6:4)

Table 21: Subject-independent evaluation configuration.

Parameter	Value
Training	89 samples
Testing	45 samples
Ratio	6:4
Key Condition	No overlap of voice samples from the same individual
Reason	Avoid biased results caused by subject overlap in cross-validation

2.5.3 Hyperparameter Tuning

Hyperparameter optimization was performed using Talos. The search space included:

- Layer sizes
- Dropout rates
- Activation functions
- Optimizers
- Learning rates
- Number of epochs
- Batch sizes

2.5.4 Software and Hardware

Table 22: Software and hardware configuration.

Component	Specification
ML Library	Scikit-learn 0.22.1
DL Library	Keras 2.2.4
GPU	Quadro M1200/PCIe/SSE2
CPU	Intel® Core™ i7-7820HQ @ 2.90 GHz × 8
OS	Ubuntu 18.04.2 LTS 64-bit
Memory	16 GiB

2.6 Evaluation Metrics

Table 23: Evaluation metrics used in the study.

Metric	Formula	Purpose
Accuracy	$\frac{TP+TN}{TP+FP+TN+FN}$	Overall correctness
Sensitivity	$\frac{TP}{TP+FN}$	Correctly identified PD cases
Specificity	$\frac{TN}{TN+FP}$	Correctly identified HC cases
F-score	$\frac{2(\text{Precision})(\text{Recall})}{\text{Precision}+\text{Recall}}$	Harmonic mean of precision and recall
MCC	Matthews Correlation Coefficient	Correlation between predicted and actual classes
Fit_time	–	Time required to fit the estimator
Score_time	–	Time required to score the estimator

2.6.1 MCC Interpretation

Table 24: Interpretation of the Matthews Correlation Coefficient.

Value	Meaning
+1	Perfect prediction
0	No better than random
-1	Complete disagreement

3 Results

3.1 Traditional ML with Static Features (10-Fold CV)

3.1.1 Input: Monophonic /a/

Table 25: Traditional ML results for monophonic /a/.

Model	Feature	Accuracy	F-score	MCC	Key Finding
SVM (Linear)	**Articulation**	**73.35%**	**79.67%**	**0.3773**	**Best overall**
MLP (500 units)	Articulation	68.19%	75.59%	0.2767	Moderate
KNN ($k = 10$)	Articulation	66.70%	74.96%	0.2254	Moderate
DT	Phonation	61.37%	69.70%	0.1304	Worst ML
GNB	Phonation	61.26%	71.60%	0.0781	Worst ML

3.1.2 Input: Short Sentence

Table 26: Traditional ML results for short sentences.

Model	Feature	Accuracy	F-score	MCC	Key Finding
DT	**Phonation**	**73.46%**	79.92%	**0.3909**	**Best overall**
SVM (RBF)	Phonation	71.10%	**80.96%**	0.2668	Best F-score
MLP (500 units)	Phonation	70.44%	76.27%	0.3563	Moderate
KNN ($k = 3$)	Phonation + Articulation	69.51%	78.86%	0.2362	Combination provides no benefit
GNB	Phonation	66.54%	70.01%	0.3388	Moderate

3.1.3 Key Observations

1. Articulation features perform better for sustained vowel detection.
2. Phonation features perform better for sentence detection.
3. Combining phonation and articulation features does not improve performance.
4. SVM performs best for the vowel task (73.35%), while DT performs best for the sentence task (73.46%).
5. No feature combination provides a consistent performance improvement.

3.2 DL Models with Dynamic Features (10-Fold CV)

3.2.1 Input: Monophonic /a/

Table 27: DL results for monophonic /a/.

Model	Features	Accuracy	F-score	MCC	Key Finding
End-to-End CNN	**Mel-scaled spectrogram**	**80.99%**	**87.00%**	**0.5319**	**Best overall**
BiLSTM	Articulation (dynamic)	77.36%	83.85%	0.4751	Good
CNN	Articulation (dynamic)	74.29%	80.75%	0.4326	Moderate

3.2.2 Input: Short Sentence

Table 28: DL results for short sentences.

Model	Features	Accuracy	F-score	MCC	Key Finding
BiLSTM	**Articulation (dynamic)**	**84.29%**	**88.52%**	**0.6603**	**Best overall**
CNN	Articulation (dynamic)	83.52%	87.48%	–	Very good
End-to-End CNN	**STFT spectrogram**	**65.24%**	**74.95%**	**0.1765**	**Worst**

3.2.3 Key Observations

1. BiLSTM with dynamic articulation features performs best for sentence detection, achieving 84.29% accuracy.
2. End-to-end CNN performs best for the vowel task, achieving 80.99% accuracy.
3. End-to-end CNN performs poorly for sentences (65.24%), suggesting that a single time-frequency representation may not sufficiently capture the complexity of continuous speech.
4. Dynamic features improve performance compared with static features.

3.3 Subject-Independent Evaluation (6:4 Split)

The subject-independent evaluation uses 89 training samples and 45 testing samples, with no overlap between voice samples belonging to the same individual.

Table 29: Subject-independent DL results.

Model	Input	Accuracy	F-score	MCC	Key Finding
BiLSTM	Sentence (dynamic)	**75.56%**	**80.70%**	**0.4811**	**Best overall**
End-to-End CNN	/a/ (Mel)	71.11%	78.69%	0.3394	Good
CNN	Sentence (dynamic)	66.67%	80.00%	0.00	Poor MCC

3.3.1 Traditional ML Baseline

Table 30: Traditional ML baseline using static articulation features for short sentences.

Model	Accuracy	F-score	MCC
SVM (poly)	64.44%	78.39%	-0.1066
KNN ($k = 5$)	60.00%	67.74%	-0.0434
MLP (200 units)	55.56%	76.27%	-0.0347
DT	48.89%	62.30%	-0.1697
GNB	44.44%	59.02%	-0.2715

3.3.2 Comparison Summary

Table 31: Comparison under subject-independent evaluation.

Approach	Best Accuracy
BiLSTM + Dynamic Features	**75.56%**
End-to-End CNN + Mel Spectrogram	71.11%
Traditional ML + Static Features	64.44%
Improvement	**+11.12%**

3.4 Hyperparameter Tuning Results (BiLSTM)

Table 32: Optimal BiLSTM hyperparameters.

Hyperparameter	**Optimal Value**
LSTM Units (Layer 1)	20
LSTM Units (Layer 2)	200
Dropout Rate	0.1
Recurrent Dropout Rate	0.1–0.2
Optimizer	Adagrad
Learning Rate	0.1
Epochs	150
Batch Size	20

The resulting configuration is:

```
Input (38 timesteps, 58 features)
        |
        v
Bidirectional LSTM
(20 units, dropout=0.1, recurrent_dropout=0.2)
        |
        v
Bidirectional LSTM
(200 units, dropout=0.1, recurrent_dropout=0.1)
        |
        v
Dense (softmax, 2 classes)
```

4 Critical Analysis

4.1 Strengths

1. **Dynamic Feature Innovation**
    - Reported first application of Bidirectional LSTM to dynamic articulation transition features.
    - Captures temporal interdependencies in speech.
    - Uses clinically motivated onset and offset transitions.

2. **Comprehensive Comparison**

   - Traditional ML using five classifiers versus DL using CNNs and RNNs.
   - Static features versus dynamic features.
   - Multiple evaluation methods.

3. **Statistical Rigor**

   - Paired t-test validation of articulation differences ($p = 0.042$).
   - Subject-independent evaluation without sample overlap.
   - Hyperparameter tuning using Talos.

4. **Multiple Speech Tasks**

   - Sustained vowel (/a/).
   - Short sentence.
   - Both tasks evaluated separately.

5. **Real-World Applicability**

   - Smartphone recordings at 96 kHz.
   - Mandarin language.
   - Hoehn & Yahr stages 1–5.

4.2 Limitations

Table 33: Limitations of the study.

Limitation	Details	Impact
Small Sample Size	45 subjects (15 HC, 30 PD)	Limited generalizability
Language-Specific	Mandarin only	Cross-lingual generalizability remains unknown
Class Imbalance	30 PD versus 15 HC (2:1 ratio)	May bias classification toward PD detection
No External Validation	Only internal cross-validation and subject-independent split	Real-world performance remains uncertain
Limited PD Stages	HY stages 1–5	No dedicated focus on early-stage PD
Simple DL Architecture	Three CNN layers and two BiLSTM layers	More complex architectures may provide improvements
No Interpretability	Black-box models	Clinical adoption may be hindered
No Foundation Model Comparison	Only conventional DL approaches were evaluated	Modern state-of-the-art baselines are absent

4.3 Comparison with Previous Work

Table 34: Comparison with previous Parkinson's disease speech-detection studies.

Study	Approach	Dataset	Acc. (%)	Ke Fin ing
Current Study	**BiLSTM + Dynamic Features**	GYENNO (Mandarin)	**75.56**	Dy fea- ture out per- forr stat fea- ture un- der sub ind eva u- a- tion
Lahmiri et al. (2017)	SVM	Oxford	92	SV per- forr str am tra- di- tion ML met ods
Saloni & Gupta (2015)	SVM	Oxford	100	Fea com bi- na- tion but po- ten- tial dat leak age
Shahbakhti et al. (2014)	GA + SVM	Oxford	94.5	Ge al- go- rith for fea- ture se-

Previous studies often reported higher accuracy, but many relied on:

- Leave-one-out cross-validation, which may allow samples from the same speaker to appear in both training and testing.
- Controlled and high-quality recordings.
- European-language datasets.
- Evaluation settings without strict subject independence.
- Smaller datasets.

Therefore, the reported 75.56% accuracy under subject-independent evaluation is competitive within the context of the study because it uses:

- A small dataset of only 45 subjects.
- Mandarin speech.
- Strict subject-independent evaluation.
- Dynamic articulation features.

4.4 Articulation Transition Analysis

The visual comparison reported in the study highlights substantial differences between HC and PD speech.

Table 35: Comparison of articulation transitions and F0 characteristics.

Aspect	**HC Speaker**	**PD Patient**
Articulation Transitions	17	39
F0 Curve	Smooth and stable	Irregular with greater variation
Statistical Significance	$p = 0.042$, indicating a significant difference	

4.4.1 Clinical Interpretation

PD patients show:

- More articulation transitions, indicating difficulty maintaining smooth articulation.
- Less stable F0 contours, indicating phonatory instability.

- Difficulty with onset and offset transitions, reflecting difficulty starting and stopping vocal fold vibration.

These observations provide motivation for using dynamic features in PD detection.

4.5 End-to-End CNN vs. Dynamic Feature Approach

Table 36: Comparison of end-to-end CNN and dynamic-feature BiLSTM approaches.

Aspect	**End-to-End CNN**	**BiLSTM + Dynamic Features**
Input	Raw spectrograms	Extracted dynamic features
Best Task	Sustained vowel (80.99%)	Short sentence (84.29%)
Subject-Independent	71.11% (vowel)	75.56% (sentence)
Interpretability	Black-box	Partially interpretable through transition-based features
Preprocessing	Spectrogram generation	Dynamic feature extraction required
Strengths	No manual feature engineering	Captures temporal dynamics

4.5.1 Hybrid Approach Suggestion

The findings suggest a possible hybrid architecture combining both approaches:

- Bidirectional LSTM with dynamic features for sentence-based detection.
- End-to-end CNN with spectrograms for vowel-based detection.
- A hybrid system could potentially improve robustness and flexibility by combining complementary information.

5 Broader Context and Related Work

5.1 Evolution of PD Speech Detection

Table 37: Evolution of machine learning approaches for PD speech detection.

Era	Approach	Key Studies	Performance
Classical ML (2008–2016)	Handcrafted features + SVM/RF	Little et al. (2008), Tsanas et al. (2012)	AUC 0.8–0.9
Feature Engineering (2014–2018)	GA, LASSO, PCA for feature selection	Shahbakhti et al. (2014), Li et al. (2017)	90–95%
Early DL (2017–2019)	CNNs on spectrograms	Vásquez-Correa et al. (2017), Wodzinski et al. (2019)	85–92%
RNN/LSTM (2019–2021)	Sequential models	Current Study	75.56% subject-independent
Foundation Models (2023–2025)	HuBERT, Wav2Vec 2.0	La Quatra et al. (2024), Favaro et al. (2023)	85–97%

5.2 RNN/LSTM for PD Detection

Table 38: RNN and related approaches for PD detection.

Study	Architecture / Features	Key Finding
Current Study	**Bidirectional LSTM + dynamic articulation**	**75.56% accuracy under subject-independent evaluation**
Wroge et al. (2018)	DNN with AVEC + GeMaps	Parkinson's diagnosis
Grover et al. (2018)	DNN with biomedical voice measures	Severity prediction (81.67%)
Kim et al. (2015)	DNN with diverse speech tasks	Severity estimation

5.2.1 Why Bidirectional LSTM?

Table 39: Advantages of Bidirectional LSTM for speech analysis.

Advantage	Explanation
Bidirectional Context	Captures information from both past and future frames.
Long-Term Memory	Overcomes the vanishing-gradient problem associated with simple RNNs.
Sequential Modeling	Preserves the temporal order of articulation transitions.
Dynamic Features	Can model the progression of speech impairment across a sequence.

5.3 Speech Tasks for PD Detection

Table 40: Comparison of speech tasks for Parkinson's disease detection.

Speech Task	Advantage	Disadvantage
Sustained Vowel	Simple, controlled, sensitive to phonation	Limited to phonatory assessment
Sentence Reading	Captures articulation and prosody; more natural	Requires literacy
DDK	Measures articulatory precision and motor speed	Non-natural speech task
Spontaneous Speech	Most natural and contains rich features	Highly variable and harder to analyze

5.3.1 Current Study Findings

Table 41: Best-performing approaches for different speech tasks.

Task	Best Approach	Accuracy
Sustained Vowel	End-to-End CNN (Mel spectrogram)	80.99%
Short Sentence	BiLSTM (dynamic articulation features)	84.29%

5.4 The Problem of Data Leakage

Data leakage can inflate reported classification performance when samples from the same speaker are present in both training and testing sets.

Table 42: Common sources of data leakage.

Practice	Problem
Leave-One-Out CV	Samples from the same speaker may appear in both training and testing.
Random Split	Multiple recordings from the same subject may be distributed across different splits.
Subject Overlap	The model may learn speaker-specific characteristics rather than disease-related characteristics.

5.4.1 Current Study's Solution

The study addresses this issue by separating the samples into training and testing sets while ensuring that all voice samples belonging to an individual are assigned exclusively to either the training or testing set.

The reported split consists of:

$$\text{Training set} = 89 \text{ samples}$$

$$\text{Testing set} = 45 \text{ samples}$$

with a 6:4 ratio and no subject overlap.

This produces lower performance than some conventional cross-validation results, but provides a more realistic estimate of generalization to unseen individuals.

6 Implications and Future Directions

6.1 Clinical Implications

6.1.1 1. Dynamic Features are Clinically Relevant

- Articulation transitions reflect underlying motor impairment.
- Difficulty with onset and offset transitions may be associated with PD.
- Dynamic measures may be sensitive to early-stage changes.

6.1.2 2. Sentence Speech is Preferable

- Sentence speech captures both phonation and articulation.
- It is more natural than sustained vowels.
- It achieved higher discriminative performance in the study (84.29% versus 77.36% for the reported BiLSTM comparisons).

6.1.3 3. Smartphone-Based Screening Feasibility

- The study uses smartphone recordings.
- No specialized recording equipment is required.
- Smartphone-based collection could support scalable remote monitoring.

6.2 Technical Implications

6.2.1 1. Dynamic Features vs. Static Features

- Temporal interdependencies are important for speech analysis.
- Bidirectional LSTM can explicitly model these dependencies.
- Dynamic features provide an improvement over traditional static-feature approaches in the reported experiments.

6.2.2 2. Task-Specific Model Selection

- For vowels: end-to-end CNN on spectrograms.
- For sentences: BiLSTM on dynamic articulation features.
- A hybrid architecture could potentially exploit both types of representations.

6.2.3 3. Subject-Independent Evaluation is Essential

- It reduces the risk of speaker-specific information leaking between training and testing.
- It provides a more realistic estimate of generalization.
- It should be considered an important evaluation standard for clinical speech classification.

6.3 Future Research Directions

Table 43: Future research directions.

Direction	Rationale	Priority
Larger Dataset	Current study is limited to 45 subjects	Critical
Multi-Lingual Validation	Test generalizability across languages	High
Early-Stage PD Focus	Focus specifically on HY stages 1–2	High
External Validation	Test on independent datasets	Critical
Hybrid Models	Combine CNN and BiLSTM representations	Medium
Multi-Task Learning	Joint detection and severity estimation	Medium
Interpretability	Explainable AI for clinical adoption	High
Multi-Modal Integration	Combine speech with gait and handwriting	Medium
Longitudinal Studies	Track disease progression over time	High

6.4 Addressing Current Limitations

Table 44: Potential solutions to the study's limitations.

Limitation	Proposed Solution
Small Sample Size	Collect a larger and balanced dataset
Class Imbalance	Balanced recruitment with approximately 1:1 PD-to-HC ratio
Language-Specific	Develop multi-lingual datasets
No External Validation	Perform cross-dataset testing
No Early-Stage Focus	Stratify evaluation according to HY stage
Black-Box Models	Apply XAI techniques such as SHAP and Grad-CAM
Simple Architecture	Investigate deeper hybrid architectures such as CNN + BiLSTM

7 Conclusion

7.1 Summary of Key Findings

1. **Articulation Transitions Differ Significantly**
    - PD patients exhibit more articulation transitions than HC speakers.
    - F0 trends are less stable in PD.
    - The reported paired t-test produced $p = 0.042$.
2. **Dynamic Features Outperform Static Features**
    - BiLSTM with dynamic articulation features achieved 84.29% accuracy under 10-fold cross-validation.
    - Traditional ML with static features achieved 73.46%.
    - Reported improvement: +10.83 percentage points.
3. **Bidirectional LSTM is Effective**
    - Captures both forward and backward temporal context.
    - Models temporal interdependencies in articulation transitions.
    - Provides the best reported performance for sentence-based detection.
4. **Task-Specific Performance**
    - Vowel: End-to-end CNN with Mel spectrogram, 80.99%.
    - Sentence: BiLSTM with dynamic features, 84.29%.
5. **Subject-Independent Evaluation Matters**
    - Provides a more realistic estimate of performance on unseen subjects.
    - BiLSTM achieved 75.56% accuracy.
    - Traditional ML achieved 64.44%.
    - Reported improvement: +11.12 percentage points.

7.2 Contributions to the Field

Table 45: Main contributions of the study.

Contribution	Significance
Dynamic Articulation Feature Innovation	Introduces onset/offset transition features for PD detection.
Bidirectional LSTM Application	Reports the first application of BiLSTM to dynamic speech features for PD detection.
Comprehensive Comparison	Compares traditional ML and DL under two evaluation methods.
Subject-Independent Validation	Addresses potential data leakage associated with subject overlap.
Speech Task Analysis	Compares vowel and sentence-based performance.
Statistical Validation	Uses a paired t-test to validate articulation differences.

7.3 Final Remarks

This study demonstrates that modeling dynamic articulation transition features with Bidirectional LSTM can improve Parkinson's disease detection compared with traditional approaches based on static speech features. The key observation that PD patients exhibit more articulation transitions and less stable F0 contours provides a clinically meaningful motivation for explicitly modeling speech dynamics.

The 75.56% accuracy achieved under subject-independent evaluation represents a more realistic benchmark for PD detection from Mandarin speech than results obtained under evaluation settings that may allow speaker overlap. Although this performance is lower than some results reported in previous studies, the evaluation protocol provides stronger control against subject-specific information leakage.

The work highlights the importance of:

- Capturing temporal dynamics in speech analysis.
- Using subject-independent evaluation for realistic performance assessment.
- Selecting task-appropriate models for different speech tasks.
- Considering hybrid approaches that combine CNN and BiLSTM representations.

Future research should address the limitations associated with small sample size, class imbalance, language specificity, and lack of external validation. Larger, multi-lingual,

early-stage-focused datasets combined with more sophisticated hybrid architectures could help determine whether dynamic speech analysis can become a clinically useful tool for non-invasive Parkinson's disease screening and monitoring.

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